

Recall, from last time,

$$\delta\tilde{\phi} = \delta\phi - \tilde{\phi}'\tau$$

But how does the metric change?

$$\begin{aligned}\text{Recall, } ds^2 &= g_{\mu\nu}(x) dx^\mu dx^\nu \\ &= \tilde{g}_{\alpha\beta}(\tilde{x}) d\tilde{x}^\alpha d\tilde{x}^\beta\end{aligned}$$

$$\text{We can write } d\tilde{x}^\alpha = \left(\frac{\partial \tilde{x}^\alpha}{\partial x^\mu}\right) dx^\mu \quad \text{and} \quad d\tilde{x}^\beta = \left(\frac{\partial \tilde{x}^\beta}{\partial x^\nu}\right) dx^\nu$$

Thus,

$$g_{\mu\nu}(x) = \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} \frac{\partial \tilde{x}^\beta}{\partial x^\nu} \tilde{g}_{\alpha\beta}(\tilde{x})$$

Defining $\tilde{x}^\mu$ as before: $\tilde{x}^\mu(q) := x^\mu(q) + \xi^\mu(q)$ where $\begin{cases} \xi^0 = \tau \\ \xi^i = L^i = \partial^i L + \hat{L}_i \end{cases}$

we can write

$$\frac{\partial \tilde{x}^\alpha}{\partial x^\mu} = \begin{pmatrix} \frac{\partial \tilde{x}^0}{\partial x^0} & \frac{\partial \tilde{x}^0}{\partial x^i} \\ \frac{\partial \tilde{x}^i}{\partial x^0} & \frac{\partial \tilde{x}^i}{\partial x^j} \end{pmatrix} = \begin{pmatrix} 1 + \tau' & \partial_j \tau \\ L^i & \delta_j^i + \partial_j L^i \end{pmatrix}$$

We label rows by α and we label columns with μ , indexing from 0

$$\alpha=0 \quad \mu=0: \frac{\partial \tilde{x}^0}{\partial x^0} = \frac{\partial \tilde{\eta}}{\partial \eta} = \frac{\partial}{\partial \eta}(\eta + \tau) = 1 + \tau'$$

$$\alpha=0 \quad \mu=j: \frac{\partial \tilde{x}^0}{\partial x^j} = \frac{\partial \tilde{\eta}}{\partial x^j} = \partial_j(\eta + \tau) = \partial_j \tau \quad (\text{Since } j \neq 0)$$

$$\alpha=i \quad \mu=0: \frac{\partial \tilde{x}^i}{\partial x^0} = \frac{\partial \tilde{x}^i}{\partial \eta} = \frac{\partial}{\partial \eta}(x^i + L^i) = (L^i)'$$

$$\begin{aligned}\alpha=i \quad \mu=j: \frac{\partial \tilde{x}^i}{\partial x^j} &= \frac{\partial}{\partial x^j}(x^i + \xi^i) \\ &= \delta_j^i + \partial_j L^i\end{aligned}$$

We now investigate what our $g_{\mu\nu} \rightarrow \tilde{g}_{\alpha\beta}$ transformation implies for the metric perturbations in our perturbed FRW metric.

P.T.O

e.g. (Temporal component) $\mu = \nu = 0$

$$g_{00}(x) = \frac{\partial \tilde{x}^\alpha}{\partial x} \frac{\partial \tilde{x}^\beta}{\partial x} \tilde{g}_{\alpha\beta}(\tilde{x})$$

Suppose that $\alpha = 0$ $\beta = i$, then the term is $\frac{\partial \tilde{x}^0}{\partial x} \frac{\partial \tilde{x}^i}{\partial x} \tilde{g}_{0i}(\tilde{x})$

$$\hookrightarrow = \frac{\partial \tilde{t}}{\partial x} \cdot L^{i'} \cdot \tilde{g}_{0i}(\tilde{x})$$

$$\propto 1 \cdot \underbrace{\xi^{i'}}_{\text{the product leads to a 2nd order perturbation}} \cdot \tilde{B}_i$$

the product leads to a 2nd order perturbation

$\Rightarrow$ it can be neglected

The process is similar for $\alpha = i$ $\beta = j$ and we conclude with the same argument.

Thus, the only contributing term is $\alpha = 0$ $\beta = 0$

$$\Rightarrow g_{00}(x) = \left(\frac{\partial \tilde{t}}{\partial x} \right) \tilde{g}_{00}(\tilde{x})$$

Recall $ds^2 = a^2(\eta) [-(1+2A)d\eta^2 + 2B_i dx^i d\eta + (\delta_{ij} + 2E_{ij}) dx^i dx^j]$

Using this and our coordinate transformation we can proceed as follows:

$$g_{00}(x) = \left(\frac{\partial \tilde{t}}{\partial x} \right)^2 \tilde{g}_{00}(\tilde{x})$$

$$= (1 + \tau')^2 \tilde{g}_{00}(\tilde{x})$$

Recall $a^2(\tilde{\eta}) = a^2(\eta + \tau)$

$$\Rightarrow a^2(\eta)(1+2A) = (1+\tau')^2 a^2(\tilde{\eta})(1+2\tilde{A})$$

$$= (1 + 2\tau' + \cancel{\tau'^2}) (a(\eta) + a'(\eta)\tau + \cancel{\dots})^2 (1 + 2\tilde{A})$$

Discarding any products higher than second order,

$$= (1 + 2\tau') (a^2(\eta) + 2a(\eta)a'(\eta)\tau) (1 + 2\tilde{A})$$

$$= a^2(\eta) (1 + 2\mathcal{H}\tau + 2\tau' + 2\tilde{A} + \dots)$$

$$= a^2(\eta) (1 + 2(\tilde{A} + \mathcal{H}\tau + \tau'))$$

Thus $\tilde{A} = A - \mathcal{H}\tau - \tau'$